

Q1

```
ec2-user@jayeshpatil:~/linuxexam
[ec2-user@jayeshpatil ~]$ mkdir linuxexam
[ec2-user@jayeshpatil ~]$ cd linuxexam
[ec2-user@jayeshpatil linuxexam]$ touch student.txt course.txt result.t
xt
[ec2-user@jayeshpatil linuxexam]$ pwd
/home/ec2-user/linuxexam
[ec2-user@jayeshpatil linuxexam]$ ls
course.txt  result.txt  student.txt
(failed reverse-i-search)`: R
```

Installed:

tree-1.8.0-6.amzn2023.0.2.x86_64

Complete!

[ec2-user@jayeshpatil linuxexam]\$ tree

```
├─ backups
│   └─ student_backup.txt
├─ documents
├─ linux_course.txt
├─ scripts
└─ student.txt
```

3 directories, 3 files

[ec2-user@jayeshpatil linuxexam]\$

```
ec2-user@jayeshpatil:~/linuxexam
[ec2-user@jayeshpatil linuxexam]$ sudo nano student.txt
[ec2-user@jayeshpatil linuxexam]$ sudo nano linux_course.txt
[ec2-user@jayeshpatil linuxexam]$ cat student.txt
1 jayesh bca
2 abhi btech
3 ashu mtech
4 Nilesh bcom
5 pravin ba
[ec2-user@jayeshpatil linuxexam]$ cat linux_course.txt
basic to advanced course
all topics will be covered in detail
6 months course
enroll
now
[ec2-user@jayeshpatil linuxexam]$ head -3 student.txt
1 jayesh bca
2 abhi btech
3 ashu mtech
[ec2-user@jayeshpatil linuxexam]$ tail -2 linux_course.txt
enroll
now
[ec2-user@jayeshpatil linuxexam]$ count student.txt
-bash: count: command not found
[ec2-user@jayeshpatil linuxexam]$ wc student.txt
 5 15 65 student.txt
[ec2-user@jayeshpatil linuxexam]$
```

```
student01@jayeshpatil:~  
[ec2-user@jayeshpatil linuxexam]$ sudo useradd -m student01  
[ec2-user@jayeshpatil linuxexam]$ sudo passwd student01  
Changing password for user student01.  
New password:  
BAD PASSWORD: The password fails the dictionary check - it is based on  
a dictionary word  
Retype new password:  
passwd: all authentication tokens updated successfully.  
[ec2-user@jayeshpatil linuxexam]$ su -u student01  
Try 'su --help' for more information.  
[ec2-user@jayeshpatil linuxexam]$ su - student01  
Password:  
[student01@jayeshpatil ~]$ exit  
logout  
[ec2-user@jayeshpatil linuxexam]$ id student01  
uid=1001(student01) gid=1001(student01) groups=1001(student01)  
[ec2-user@jayeshpatil linuxexam]$ pwd  
/home/ec2-user/linuxexam  
[ec2-user@jayeshpatil linuxexam]$ cd ~  
[ec2-user@jayeshpatil ~]$ getent passwd student01  
student01:x:1001:1001::/home/student01:/bin/bash  
[ec2-user@jayeshpatil ~]$ su - student01  
Password:  
su: Authentication failure  
[ec2-user@jayeshpatil ~]$ su - student01  
Password:  
Last login: Tue Aug 25 10:00:22 UTC 2026 on pts/2  
Last failed login: Tue Aug 25 10:04:55 UTC 2026 on pts/2  
There was 1 failed login attempt since the last successful login.  
[student01@jayeshpatil ~]$ |
```

```
ec2-user@jayeshpatil:~  
[ec2-user@jayeshpatil ~]$ sudo groupadd linuxbatch  
[ec2-user@jayeshpatil ~]$ sudo usermod -aG linuxbatch student01  
[ec2-user@jayeshpatil ~]$ sudo useradd student02  
[ec2-user@jayeshpatil ~]$ sudo usermod -aG linuxbatch student02  
[ec2-user@jayeshpatil ~]$ grep linuxbatch /etc/group  
linuxbatch:x:1002:student01,student02  
[ec2-user@jayeshpatil ~]$
```

```
ec2-user@jayeshpatil:~/linuxexam
[ec2-user@jayeshpatil ~]$ cd linuxexam
[ec2-user@jayeshpatil linuxexam]$ touch project.txt
[ec2-user@jayeshpatil linuxexam]$ chmod 754 project.txt
[ec2-user@jayeshpatil linuxexam]$ ls -l
total 8
drwxr-xr-x. 2 ec2-user ec2-user 32 Aug 25 09:42 backups
drwxr-xr-x. 2 ec2-user ec2-user  6 Aug 25 09:41 documents
-rw-r--r--. 1 ec2-user ec2-user 90 Aug 25 09:51 linux_course.txt
-rwxr-xr--. 1 ec2-user ec2-user  0 Aug 25 10:16 project.txt
drwxr-xr-x. 2 ec2-user ec2-user  6 Aug 25 09:41 scripts
-rw-r--r--. 1 ec2-user ec2-user 65 Aug 25 09:47 student.txt
[ec2-user@jayeshpatil linuxexam]$ chmod 640 project.txt
[ec2-user@jayeshpatil linuxexam]$ sudo chown student01 project.txt
[ec2-user@jayeshpatil linuxexam]$ sudo chgrp linuxbatch project.txt
[ec2-user@jayeshpatil linuxexam]$ ls -l project.txt
-rw-r-----. 1 student01 linuxbatch 0 Aug 25 10:16 project.txt
[ec2-user@jayeshpatil linuxexam]$
```

```
ec2-user@jayeshpatil:~/linuxexam
[ec2-user@jayeshpatil linuxexam]$ mkdir public private shared
[ec2-user@jayeshpatil linuxexam]$ chmod 777 public
[ec2-user@jayeshpatil linuxexam]$ chmod 700 private
[ec2-user@jayeshpatil linuxexam]$ chmod 770 shared
[ec2-user@jayeshpatil linuxexam]$ ls -ld
drwxr-xr-x. 8 ec2-user ec2-user 158 Aug 25 10:21 .
[ec2-user@jayeshpatil linuxexam]$ ls -ld public
drwxrwxrwx. 2 ec2-user ec2-user 6 Aug 25 10:21 public
[ec2-user@jayeshpatil linuxexam]$ ls -ld private
drwx-----. 2 ec2-user ec2-user 6 Aug 25 10:21 private
[ec2-user@jayeshpatil linuxexam]$ ls -ld shared
drwxrwx---. 2 ec2-user ec2-user 6 Aug 25 10:21 shared
[ec2-user@jayeshpatil linuxexam]$
```

```
ec2-user@jayeshpatil:~$ sudo yum update -y
Last metadata expiration check: 1:40:05 ago on Tue Aug 25 08:51:32 2026
Dependencies resolved.
Nothing to do.
Complete!
[ec2-user@jayeshpatil ~]$ sudo yum install httpd -y
Last metadata expiration check: 1:40:13 ago on Tue Aug 25 08:51:32 2026
Package httpd-2.4.68-1.amzn2023.0.1.x86_64 is already installed.
Dependencies resolved.
Nothing to do.
Complete!
[ec2-user@jayeshpatil ~]$ httpd -v
Server version: Apache/2.4.68 (Amazon Linux)
Server built:   Jun  8 2026 00:00:00
[ec2-user@jayeshpatil ~]$ sudo service httpd start
Redirecting to /bin/systemctl start httpd.service
[ec2-user@jayeshpatil ~]$ sudo service httpd status
Redirecting to /bin/systemctl status httpd.service
● httpd.service - The Apache HTTP Server
   Loaded: loaded (/usr/lib/systemd/system/httpd.service; disabled; vendor preset: enabled)
   Drop-In: /usr/lib/systemd/system/httpd.service.d
            └─php-fpm.conf
   Active: active (running) since Tue 2026-08-25 10:32:24 UTC; 1min ago
     Docs: man:httpd.service(8)
  Main PID: 7847 (httpd)
    Status: "Total requests: 0; Idle/Busy workers 100/0; Requests/sec: 0.00"
     Tasks: 177 (limit: 1059)
    Memory: 13.7M
       CPU: 149ms
    CGroup: /system.slice/httpd.service
            └─7847 /usr/sbin/httpd -DFOREGROUND
              7848 /usr/sbin/httpd -DFOREGROUND
              7849 /usr/sbin/httpd -DFOREGROUND
              7853 /usr/sbin/httpd -DFOREGROUND
              7862 /usr/sbin/httpd -DFOREGROUND

Aug 25 10:32:24 jayeshpatil systemd[1]: Starting httpd.service - The Apache HTTP Server:
Aug 25 10:32:24 jayeshpatil httpd[7847]: AH00558: httpd: Could not reliably determine the
Aug 25 10:32:24 jayeshpatil systemd[1]: Started httpd.service - The Apache HTTP Server:
Aug 25 10:32:24 jayeshpatil httpd[7847]: Server configured, listening on: 0.0.0.0:80

[ec2-user@jayeshpatil ~]$ sudo systemctl enable httpd
```



```
ec2-user@jayeshpatil:/var/www/html
[ec2-user@jayeshpatil html]$ cd /var/www/html
[ec2-user@jayeshpatil html]$ sudo nano index.html
[ec2-user@jayeshpatil html]$ sudo service httpd restart
Redirecting to /bin/systemctl restart httpd.service
[ec2-user@jayeshpatil html]$
```

jayesh patil 21 devops "linux practical examinations"

```

ec2-user@jayeshpatil:/var/www/html
Process: 8560 ExecStart=/usr/sbin/httpd $OPTIONS -DFOREGROUND (code=exited, status=0/SUCCESS)
Main PID: 8560 (code=exited, status=0/SUCCESS)
Status: "Total requests: 2; Idle/Busy workers 100/0;Requests/sec: 2.00"
CPU: 522ms

Aug 25 10:40:54 jayeshpatil systemd[1]: Starting httpd.service - The Apache HTTP Server
Aug 25 10:40:54 jayeshpatil httpd[8560]: AH00558: httpd: Could not rel
Aug 25 10:40:54 jayeshpatil systemd[1]: Started httpd.service - The Apache HTTP Server
Aug 25 10:40:54 jayeshpatil httpd[8560]: Server configured, listening on
Aug 25 10:47:26 jayeshpatil systemd[1]: Stopping httpd.service - The Apache HTTP Server
Aug 25 10:47:27 jayeshpatil systemd[1]: httpd.service: Deactivated successfully
Aug 25 10:47:27 jayeshpatil systemd[1]: Stopped httpd.service - The Apache HTTP Server

[ec2-user@jayeshpatil html]$ sudo service httpd start
Redirecting to /bin/systemctl start httpd.service
[ec2-user@jayeshpatil html]$ sudo service httpd start
Redirecting to /bin/systemctl start httpd.service
[ec2-user@jayeshpatil html]$ sudo service httpd status
Redirecting to /bin/systemctl status httpd.service
● httpd.service - The Apache HTTP Server
   Loaded: loaded (/usr/lib/systemd/system/httpd.service; enabled; preset=yes)
   Drop-In: /usr/lib/systemd/system/httpd.service.d
            └─php-fpm.conf
   Active: active (running) since Tue 2026-08-25 10:49:11 UTC; 18s ago
     Docs: man:httpd.service(8)
  Main PID: 9302 (httpd)
    Status: "Total requests: 0; Idle/Busy workers 100/0;Requests/sec: 0.00"
     Tasks: 177 (limit: 1059)
    Memory: 13.7M
       CPU: 78ms
    CGroup: /system.slice/httpd.service
            └─9302 /usr/sbin/httpd -DFOREGROUND
              └─9303 /usr/sbin/httpd -DFOREGROUND
                └─9304 /usr/sbin/httpd -DFOREGROUND
                  └─9305 /usr/sbin/httpd -DFOREGROUND
                    └─9414 /usr/sbin/httpd -DFOREGROUND

Aug 25 10:49:11 jayeshpatil systemd[1]: Starting httpd.service - The Apache HTTP Server
Aug 25 10:49:11 jayeshpatil httpd[9302]: AH00558: httpd: Could not rel
Aug 25 10:49:11 jayeshpatil systemd[1]: Started httpd.service - The Apache HTTP Server
Aug 25 10:49:11 jayeshpatil httpd[9302]: Server configured, listening on

[ec2-user@jayeshpatil html]$ ^C

```

Q12

```
ec2-user@jayeshpatil:~/linuxexam
[ec2-user@jayeshpatil linuxexam]$ nano students.txt
[ec2-user@jayeshpatil linuxexam]$ grep "a" students.txt
101 Jayesh Patil
102 Rahul Sharma
104 Sneha Patil
105 Priya Shah
106 Akash More
107 Neha Kulkarni
108 Rohan Deshmukh
109 Anjali Pawar
110 Karan Jadhav
[ec2-user@jayeshpatil linuxexam]$ grep -i "a" students.txt
101 Jayesh Patil
102 Rahul Sharma
103 Amit Joshi
104 Sneha Patil
105 Priya Shah
106 Akash More
107 Neha Kulkarni
108 Rohan Deshmukh
109 Anjali Pawar
110 Karan Jadhav
[ec2-user@jayeshpatil linuxexam]$ grep -ic "a" students.txt
10
[ec2-user@jayeshpatil linuxexam]$ sort students.txt
101 Jayesh Patil
102 Rahul Sharma
103 Amit Joshi
104 Sneha Patil
105 Priya Shah
106 Akash More
107 Neha Kulkarni
108 Rohan Deshmukh
109 Anjali Pawar
110 Karan Jadhav
[ec2-user@jayeshpatil linuxexam]$ find . -name "students.txt"
./students.txt
[ec2-user@jayeshpatil linuxexam]$ find . -name "*.txt"
./student.txt
./linux_course.txt
./backups/student_backup.txt
./project.txt
./students.txt
[ec2-user@jayeshpatil linuxexam]$
```

```
ec2-user@jayeshpatil:~/linuxexam
[ec2-user@jayeshpatil linuxexam]$ hostnamectl
Static hostname: jayeshpatil
Icon name: computer-vm
Chassis: vm
Machine ID: ec2fb5d744dd4ba1f9ab3d6baecafe6b
Boot ID: 602e3262b79b494282acf7a8523298ad
Virtualization: amazon
Operating System: Amazon Linux 2023.12.20260817
CPE OS Name: cpe:2.3:o:amazon:amazon_linux:2023
Kernel: Linux 6.18.41-94.142.amzn2023.x86_64
Architecture: x86-64
Hardware Vendor: Amazon EC2
Hardware Model: t3.micro
Firmware Version: 1.0
[ec2-user@jayeshpatil linuxexam]$ hostname -I
172.31.3.227
[ec2-user@jayeshpatil linuxexam]$ ip link
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN mode DEFAULT group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
2: ens5: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 9001 qdisc mq state UP mode DEFAULT group default qlen 1000
    link/ether 02:6f:8e:63:97:0d brd ff:ff:ff:ff:ff:ff
    altname enp0s5
    altname eni-0a51bf3ca8ff23f8c
    altname device-number-0.0
[ec2-user@jayeshpatil linuxexam]$ ip route
default via 172.31.0.1 dev ens5 proto dhcp src 172.31.3.227 metric 512
172.31.0.0/20 dev ens5 proto kernel scope link src 172.31.3.227 metric 512
172.31.0.1 dev ens5 proto dhcp scope link src 172.31.3.227 metric 512
172.31.0.2 dev ens5 proto dhcp scope link src 172.31.3.227 metric 512
[ec2-user@jayeshpatil linuxexam]$ ping google.com
PING google.com (142.251.179.101) 56(84) bytes of data:
64 bytes from pd-in-f101.1e100.net (142.251.179.101): icmp_seq=1 ttl=10
6 time=2.62 ms
64 bytes from pd-in-f101.1e100.net (142.251.179.101): icmp_seq=2 ttl=10
6 time=2.61 ms
64 bytes from pd-in-f101.1e100.net (142.251.179.101): icmp_seq=3 ttl=10
6 time=2.61 ms
```